

# CHUA HENG LI

Engineering Product Development @ SUTD · Electrical  
Track

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YOUR  
PHOTO

Hardware · Embedded · Data — I design, simulate and build across the electrical-engineering stack: RF & antenna simulation in CST, embedded firmware on ESP32-S3 & Arduino, and end-to-end data pipelines in Python.

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## ABOUT

# Engineering that moves people, data and signals

I am a 2nd-year Engineering Product Development undergraduate at SUTD, specialising in the Electrical Track. My work sits at the intersection of hardware, software and data: I prototype embedded systems, simulate RF & antenna designs, and build production-style data pipelines in Python.

I've taken a student team to 1st Prize at SEMI Tech Zoomers 2026 (Southeast Asia), designed a motor-subsystem for a medical assistive device, laid out a custom PCB and wrote the firmware for an industrial robot, and built a full stock-portfolio analytics stack on Moomoo's OpenD API.

I'm drawn to problems where careful engineering analysis removes guesswork — from Monte-Carlo sizing to CST full-wave simulation.

Chua Heng Li · Portfolio

### 1st Prize

SEMI Tech Zoomers 2026

### 4.2 / 5.0

cGPA · Global Merit Scholar

### 1,000,000

Monte-Carlo sims · motor sizing

### 2.45 GHz

CST antenna designs

## SELECTED PROJECTS

# From full-wave RF to embedded robotics

### DATA & FINANCE

## Stock Portfolio Tracker

Python · Moomoo OpenD API · SQLite · Streamlit

### 3D PROJECT

## 'Alleviate' Sit-to-Stand Device

ST Engineering For Good · Monte-Carlo sizing · Arduino

### RF / EM

## Flexible Antenna — CST

2.45 GHz · S11, efficiency, farfield · (also stretchable BLE tape)

### 30.007 EDI

## Arm-Subsystem Robot

KiCad PCB · ESP32-S3 firmware · industrial material flow

# Stock Portfolio Tracker

*Personal Project · Nov 2025 – Feb 2026*

**Automated data pipeline** — Python integrates Moomoo's OpenD gateway → daily portfolio retrieval & storage

**SQLite schema** — persists positions, transactions, and historical daily snapshots for querying

**Streamlit + Plotly dashboard** — benchmarks returns vs major world indices, auto-refresh, trade ledger

**Finance-grade analytics** — Time-Weighted Return from inception; classifies dividends/coupons vs external capital

**Quality & security** — pytest unit tests; credentials isolated outside the repo

**DASHBOARD  
SCREENSHOT**

→ add image here

# ‘Alleviate’ Sit-to-Stand Bed Assist

*SUTD 3D Project · team of 5 · Jan – May 2026*

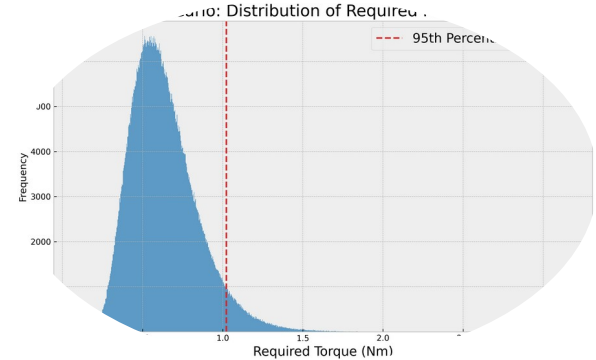
**Problem:** help elderly with limited strength transition bed → standing, independently

**Approach:** low-cost modular add-on to existing beds; mechanical rail + sliding mat

**My subsystem (CaDDE):** modelled motor torque with a 1,000,000-run Monte-Carlo (user mass, friction, shaft-distribution)

**Result:**  $\geq 1.026$  Nm to assist 95% of target demographic; safety factor 2 → 2.052 Nm across dual 12V motors

**Built with:** Arduino Uno, motor drivers, limit switches + emergency stop, 3D-printed rails



*Monte-Carlo required-torque model — my contribution*

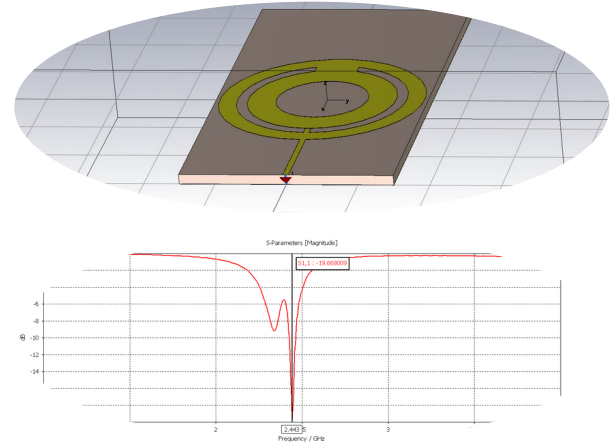
# Flexible Antenna Design in CST

*SUTD Electromagnetics & Applications · Jul 2026*

**Concentric-circle flexible antenna:** designed & simulated for the 2.45 GHz ISM band — S11 match, VSWR, radiation & total efficiency, farfield directivity/gain in CST Studio Suite

**Stretchable BLE ‘Kintenna’ antenna (exploration):** 2.44 GHz liquid-metal (EGaIn) dipole on kinesiology tape — engineered for strain tolerance (full-wave SAFe figures: reduces drift ~25% vs a flat control at 20% strain)

**Automation:** parametric CST VBA macro generation for rapid antenna iterations



*3D view + S11 (return loss)*

# Autonomous Industrial Material-Flow Robot

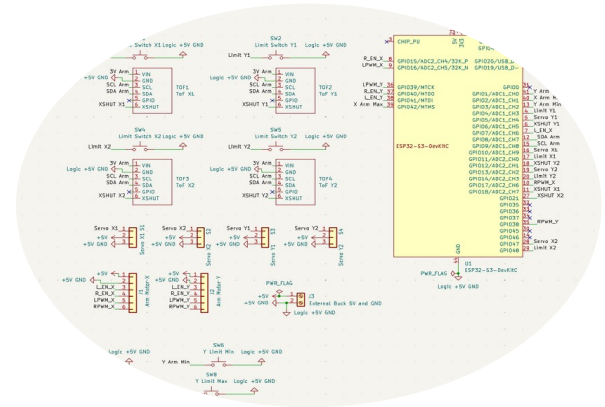
*Team project · System Design Review → public showcase Aug 2026*

**Concept:** a low-profile, flat form-factor wheeled robot that augments warehouse material flow at a textile factory — moves non-standard cage trolleys (~70 kg demo) with no floor/ trolley retrofits

**My contribution — Arm subsystem:** designed the arm-subsystem PCB in KiCad (schematic → layout)

**Embedded firmware:** Arduino/PlatformIO on ESP32-S3: ToF sensor I<sup>2</sup>C addressing via XSHUT, servo/motor control, limit-switch homing, and a Homing→Extending→Pre-flip→Flipping→Retracting state machine with safety stops

**Electronics:** prototyped & validated ESP32-S3, ToF, servos, BTS7960 motor-driver interfaces



*Arm-subsystem PCB — KiCad (my design)*

# Wins, tools and where I've worked

## Awards & Recognition

### 1st Prize — SEMI Tech Zoomers 2026

SEA industry comp, mentored by SSMC

### Global Merit Scholarship

SUTD 2024

### Sustainability Practice Prize

DTI Bird Deterrent 2025

## Skills & Tools

**Languages:** Python, C, C++

**Data / ML:** pandas, NumPy, Plotly, Streamlit, scikit-learn

**Hardware:** Arduino, ESP32, KiCad, BTS7960, sensors

**CAD / Simulation:** AutoCAD, Fusion360, CST Studio Suite

**Dev:** Git, GitHub, Docker

**Languages:** English · Mandarin

## Experience

### UOB Kay Hian

Operations Assistant · Mar 24 – Sep 24  
Dividend & corporate-action data reconciliation



1st Prize certificate